

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Full stack engineer with production TypeScript and Node.js experience in healthcare-adjacent AI systems. At Isomer AI, built a claims automation platform end-to-end: multi-model LLM routing (Claude, GPT-4o, Gemini), real-time event pipelines, document processing, and workflow orchestration - 394 commits and 7 production deployments in a small team. Built a privacy-first RAG system for healthcare and fintech data environments. Healthcare domain depth from a health ML app ranked top 1% of 1,100+ submissions and a health claim automation tool that improved accuracy by 25%. MS CS at NC State (May 2026). Portfolio: kulkarniaditya.com

CORE COMPETENCIES

TypeScript / Node.js

React / Next.js

PostgreSQL / Redis

REST APIs

LLM Pipelines (Claude, GPT-4o, Gemini)

MCP Tooling

RAG / Healthcare AI

Temporal.io Workflows

AWS / GCP

Docker / CI/CD

Observability (Sentry, Prometheus, OTel)

Full Stack Ownership

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Shipped 10+ backend features across 7 production deployments for an AI-powered claims automation platform; 394 commits in TypeScript, Node.js, Express, PostgreSQL, Redis, and Prisma with real-time event pipelines via Pusher and observability via Sentry
- Built multi-model LLM routing across Claude, GPT-4o, and Gemini (8+ models) via MCP tooling; implemented Temporal.io workflow orchestration, RBAC, document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), and Auth0 authentication

NC State University April 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform (JBrowse 2 on Google Cloud Run) serving 500GB datasets to 100+ researchers worldwide; improved ML model accuracy by 12% via hyperparameter tuning

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built a health claim automation tool using OCR, NLP, and ML -- improved claim processing accuracy by 25%; automated deployments on AWS and GCP with Grafana monitoring dashboards

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed pipelines (Hadoop + Spark batch, Kafka streaming) for support ticket automation; reduced processing time by 15% and latency by 20%

PROJECTS

Glance AI - Privacy-First RAG System Healthcare / Fintech AI

Per-tenant data isolation at the vector retrieval layer for healthcare and fintech environments where data leakage is a compliance risk. Each retrieval context is scoped to its tenant boundary before LLM input. Deterministic output architecture ensures factual correctness for regulated data -- no hallucinations in sensitive patient or financial contexts.

Python, LLMs, Vector Databases, RAG

[AlgoPulse - Agentic AI Platform](#)

Event-Driven / Scale

Identified API-polling bottleneck under inference load; re-architected to event-driven streaming with Supabase Edge Functions. Temporal.io for durable workflow state; Kafka for fault-tolerant worker coordination. Reached 2,000 async inference requests/second with zero message loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

[PulseRank - Real-Time Leaderboard & Feed](#)

Distributed Systems

Multi-store event-driven platform: MySQL outbox + idempotency keys, MongoDB TTL projections, Redis sorted sets for sub-ms reads; live updates over Server-Sent Events. Multi-store consistency validated with Testcontainers; benchmarked with k6; OpenTelemetry + Prometheus observability.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

[LLM Eval Harness](#)

AI/ML Ops

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on a 2,000-example frozen test set. Tracked hallucination rate, p50/p95 latency, and cost/call per model with statistical significance. DuckDB + Parquet analytics; Streamlit dashboard.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

[Sanjeevani - Predictive Health App](#)

Top 1% / Healthcare ML

ML-powered health app achieving 87% classification accuracy on real-time sensor data. Ranked top 1% of 1,100+ submissions (top 12). Recognized for UI innovation and predictive health insights; full-stack build with React.

React, ML classification models, real-time sensor data

[PerCent - GenAI Finance App](#)

HackNC24 Top 10

Complete RAG-based GenAI finance app built in 48 hours: goal tracking, expense logging, AI chatbot using Qwen-14B-Chat. Top 10 of 100+ teams; reduced manual financial research by 30%.

Python, Hugging Face, Qwen-14B-Chat, RAG, Selenium

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University -- BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 Machine Learning, Big Data, DS & Algorithms, Blockchain	

AWARDS & LEADERSHIP

Sanjeevani Health App -- Top 1% of 1,100+ submissions (top 12); 87% ML accuracy	2023
HackNC24 Top 10 -- GenAI Finance App; top 10 of 100+ teams in 48-hour build	2024
IEEE Publication -- Mapify: T5 pipeline, 60% review time reduction, 500+ attendees	2022
Google Developer Student Club Lead -- 4+ events, 500+ students	2022-2024

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, JavaScript, SQL, R, C	Frontend: React, Next.js, Tailwind CSS
Backend: Node.js, Express, NestJS, FastAPI, Spring Boot, REST APIs	
AI/ML: MCP, LangChain, Hugging Face, PyTorch, RAG, OpenAI/Anthropic/Google SDKs	
Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant	Infrastructure: AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, Terraform
Observability: Sentry, Prometheus, Grafana, OpenTelemetry	
DevOps: GitHub Actions, Docker Compose, Testcontainers	